Sabancı University

Spring 2025–2026

CS 412 – Machine Learning

Course Information

Level of Course	Undergraduate
Prerequisites	**(MATH 201 and MATH 203)**
Course Page	https://sucourse.sabanciuniv.edu/plus/course/view.php?id=12843

Time & Place

Time	Week Day	Place
16:40–17:30	Monday	SBS-G071
16:40–18:30	Tuesday	UC-G030

Instructor Information

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Course Description

This is an introductory machine learning course that will aim a solid understanding of the fundamental issues in machine learning (overfitting, bias/variance), together with several state-of-art approaches such as decision trees, linear regression, k-nearest neighbor, Bayesian classifiers, support vector machines, neural networks, logistic regression, and classifier combination.

Course Learning Outcomes

1. Have a solid understanding of the basic concepts, issues, assumptions and limitations in machine learning and how they apply to various machine learning techniques.
2. Have a working knowledge of the basic mathematics (probability, expectation, entropy, basic linear algebra, ...) and algorithms behind common machine learning techniques; together with their suitability in given situations.
3. Given a machine learning problem, select, implement and evaluate one of the appropriate machine learning algorithms (e.g. decision trees, neural networks, SVMs) using a programming environment such as Weka or Matlab.

Course Objective

To teach fundamentals of Machine Learning to students of all backgrounds, so that they will know its capabilities and limitations, and be able to design all aspects of a learning system.

Learning Objectives

1. Understand the basic concepts, issues, assumptions, and limitations in machine learning (e.g., base accuracy, overfitting, bias/variance, curse of dimensionality...).
2. Have a working knowledge of the basic mathematics (probability, expectation, entropy, basic linear algebra, ...) and algorithms behind common machine learning techniques, together with their suitability in given situations.
3. Given a machine learning problem, be able to implement and evaluate one of the standard machine learning algorithms (e.g., decision trees, neural networks, SVMs) using a tool such as Weka/Matlab or a programming language such as Python/R.

Course Materials

Textbook

- **None.** Normally, slides and lectures will be sufficient to do well on the exam; however, references to sections in one or two of the reference books below will be provided as a supplement to the lecture slides.
- Note that many Machine Learning or Pattern Recognition books are available online by their authors, and others will be on reserve in the IC.

Reference Books

- The Elements of Statistical Learning – Hastie et al. (freely available)
- Probabilistic Machine Learning: An Introduction – K. Murphy (free draft version)
- Machine Learning – Ethem Alpaydin (online book available through IC)
- Neural Networks and Learning Machines – Haykin (e-book)
- Pattern Recognition and Machine Learning – Bishop (no online version)

Assessment

Assessment Method	Percentage (%)
Final	**35%**
Midterm	**30%**
Quizzes	10%
Homeworks	25%

Quizzes

- **No makeup quizzes** will be given for the short quizzes.
- We will have a quiz in almost every lecture hour via TopHat, and each quiz grade will be calculated as 50% attendance and 50% correctness.
- The final quiz grade will be the average of the best $n - 3$ quiz scores, where n is the total number of quizzes.
- Please note that the number of quizzes stated above is an estimate provided to satisfy form requirements; the actual number of quizzes may differ.

Homeworks

There will be **five homework assignments** in total, with no separate project.

Passing Grade

> To pass the course, your overall grade as calculated above must be at least **40/100 (strict)**, and your final exam grade must be above **29/100** (i.e., you should be able to answer approximately one-third of the questions to demonstrate that you have learned the material).

Policies

Time Conflicts & Attendance

- Time conflicts will be approved only with the understanding that students take full responsibility for any missed lectures and any resulting missed quiz grades.
- **Regular lecture attendance is strongly encouraged** for effective learning and for achieving a good course grade.

Use of AI

- You may use ChatGPT or similar tools to obtain limited programming assistance for homework assignments.
- You are strongly discouraged from using such tools for "thinking" or "knowledge-based" questions, as similar questions may appear on exams.
- Any LLM usage must be for partial help only, not for generating complete answers.
- Homework submissions in which the code and report appear inconsistent or suggest overreliance on LLM-generated content may receive a grade of zero.
- **The use of LLMs or any AI tools during quizzes or exams is strictly prohibited.** Doing so constitutes a violation of academic integrity and will result in disciplinary action if detected.

Make-Up

- No make-up quizzes will be given for short quizzes.
- If you miss a midterm or the final exam due to a documented valid excuse (e.g., a medical report), you will be allowed to take a make-up exam, which will take place after the final exams.

Late Submission

Late homework will be accepted for up to two days, with a penalty of **10 points per late day**. Submissions made within the first hour after the deadline will receive a reduced penalty of **5 points**.

Tentative Outline

1. Week 1-2: Introduction to ML concepts: supervised learning, regression, classification, features, train-validation, overfitting, error measures, ...
2. Week 3-4: Simple ML approaches: Linear regression; Logistic regression; Nearest Neighbor Classifier; Decision tree learning; Bias and Variance
3. Week 5-6: Bayesian Approaches: Bayes theorem, Naïve Bayes, Multivariate Gaussian distribution, Maximum-likelihood estimation
4. Week 7-8: Neural Networks – Multi-Layer Perceptron, Activation functions, Capabilities, Backpropagation
5. Week 9-10: Ensemble Methods
6. Week 11-12: Deep learning: Convolutional neural Networks, Transfer Learning
7. Week 13-14: Unsupervised learning